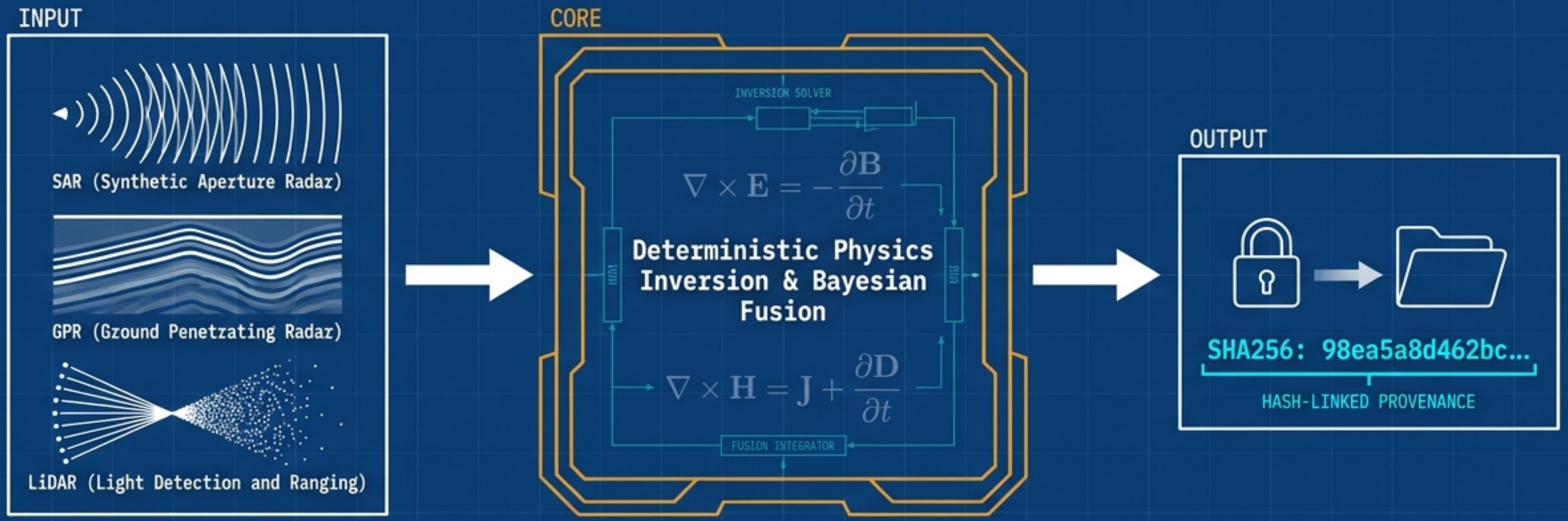
Cryptographically Secured Multimodal Physics Inversion Engine for Forensic Metrology (PHYS-MET / HFMR)



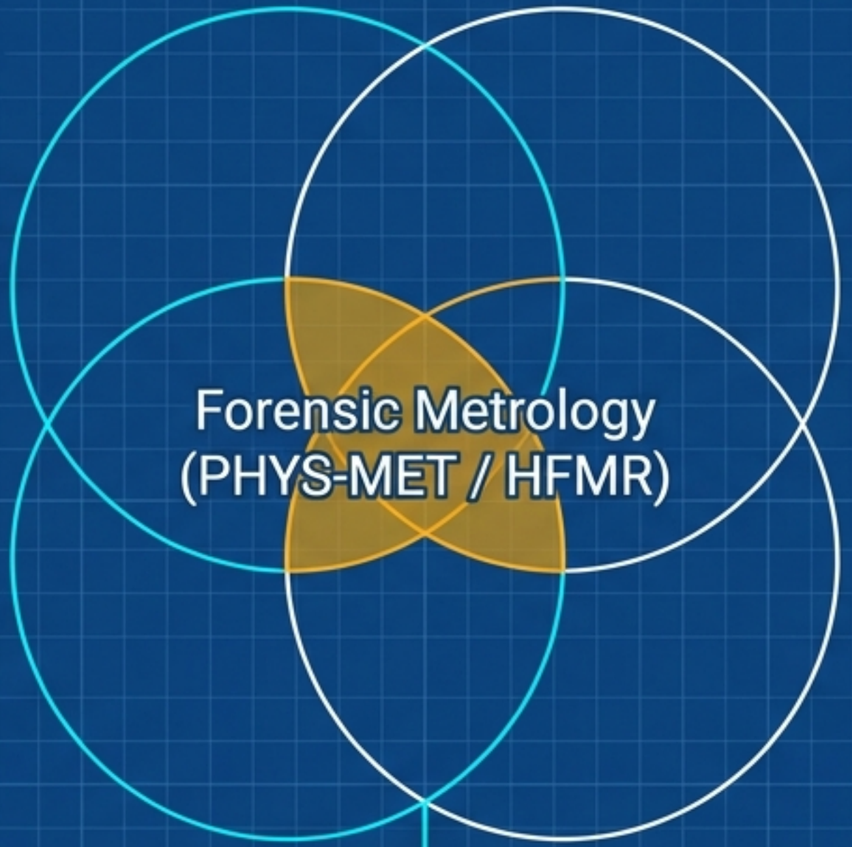
- An algorithmic framework and hardware subsystem for detecting, mapping, classifying, and documenting anomalous objects and subsurface matrices.
- Bypasses subjective AI classification by directly measuring physical properties, comparing against known material calibration standards, and outputting mathematically defensible, hash-linked evidentiary provenance.

Gemini Notebook

Defining the operational domain of secure empirical physics metrology

Deterministic Physics
(Maxwell-based wave propagation, optical interferometry)

Geospatial Intelligence
(SAR backprojection, bare-earth LiDAR)



Materials Science
(Dielectric attenuation, XRF elemental signatures)

Cryptography
(Immutable DAG ledgers, Ed25519 signatures)

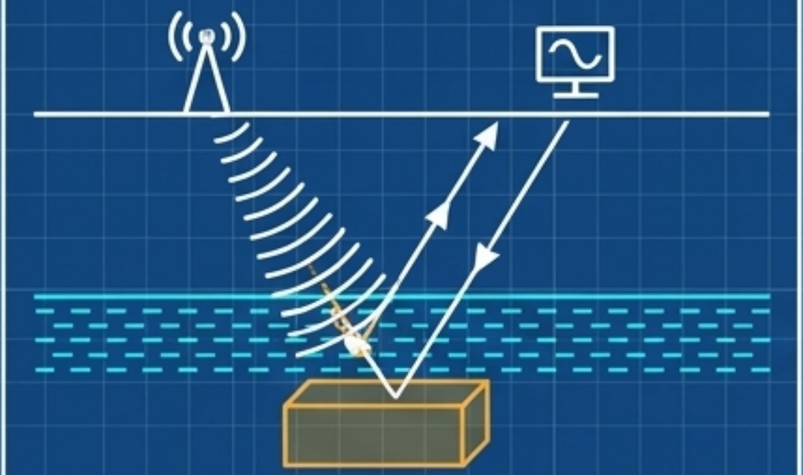
- The system intersects physical waveform reconstruction with immutable cryptographic auditing.
- Establishes a novel field focused on post-acquisition forensic validation rather than real-time autonomous navigation or visual estimation.

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PROVISIONAL APP. / DOCKET SEC. 3: BACKGROUND & PRIOR ART LIMITATIONS

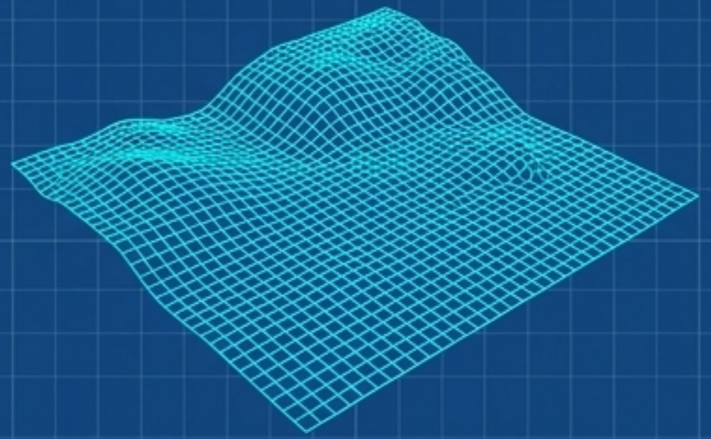
Isolated sensor systems fail to provide scientifically defensible material certainty

Standard SAR/GPR



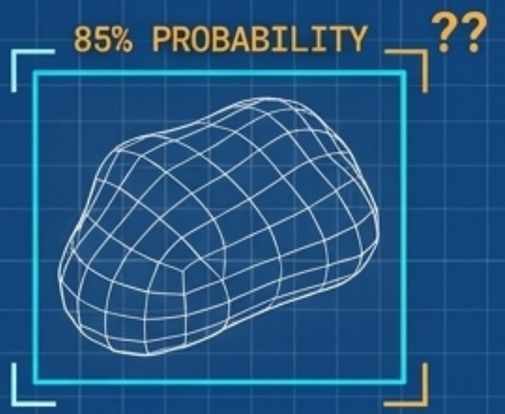
FAILURE POINT: Cannot reliably identify elemental composition from shape or reflectivity alone; suffers from soil attenuation ambiguity.

Optical/LiDAR Metrology



FAILURE POINT: Precise geometry, but cannot penetrate ground cover or determine subsurface material makeup.

Neural Network / AI Guessing



FAILURE POINT: Lacks scientific defensibility; black-box algorithms cannot report deterministic physical parameter uncertainty for court use.

- Current methodologies conflate detection (presence of an anomaly) with identification (elemental composition).
- A reliable system must strictly separate these outputs and report unmodeled effects.

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Bypassing existing drone navigation and trigger trigger patents via deterministic inversion

PROVISIONAL APP. / DOCKET SEC. 4: NOVELTY DIFFERENTIATION MATRIX

Patent/System	Claim	Differentiation
US9915945B2 (IBM)	Drone launches based on biosensor/GPS triggers to locate individuals.	PHYS-MET Differentiation: We claim ZERO flight operations, autonomous launch triggers, or biological monitoring. PHYS-MET is a static data-processing engine applied to payload waveforms.
US12743734 (LiDAR Drone)	Aerial navigation combining LiDAR point-clouds and computer vision to track cell signals.	PHYS-MET Differentiation: We do not use real-time computer vision for tracking. We claim offline deterministic physics inversion to reconstruct material properties, explicitly discarding visual pixel data.
Proposed Architecture (PHYS-MET)	Hash-linked material hypothesis testing.	Result: Absolute non-overlap. The engine establishes cryptographic provenance of simulated electromagnetic responses vs. raw data, independent of the acquisition vehicle.

[PATENT PENDING] / CONFIDENTIAL DISCLOSURE

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A multimodal physics inversion pipeline outputting court-ready evidence

Ingestion: Raw phase, amplitude, frequency, and geometry from historical or active sensors (SAR, GPR, LiDAR).

Inversion Engine: Evaluates competing hypotheses (Aluminum, Brass, Void, Soil). Simulates expected electromagnetic response under actual sensor configurations ($\hat{y}_m = F(m, \theta)$).

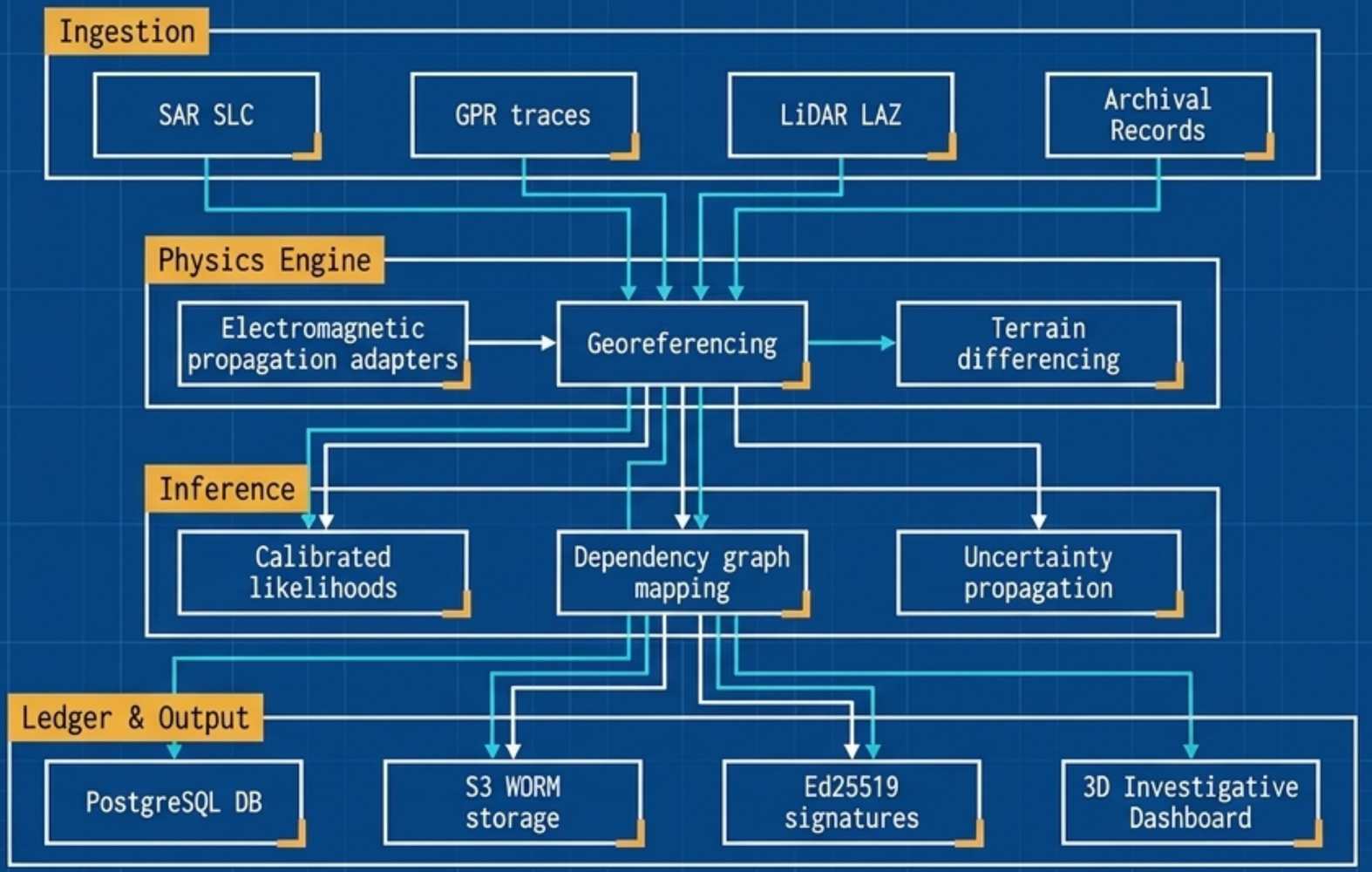
Authentication: Produces a weighted residual of the best fit alongside parameter uncertainty, anchoring the result to a SHA-t to a SHA-256 evidence log.

- Calculates a mathematically defensible material identification.
- Produces two separate, cryptographically signed outputs: anomaly detection confidence and elemental identification probability.

[PATENT PENDING]
CONFIDENTIAL DISCLOSURE

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Routing logic for the historical forensic material reanalysis engine (HFMR)



Designed as a retrospective discovery engine to reprocess historical datasets, with modular inputs for real-time dual-comb optical arrays.

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Mathematical reconstruction and hypothesis testing of physical properties

Step 1
Dual-comb interferometry frequencies:

$$f_n = f_{CEO} + n f_{rep}$$

Step 2
Time-of-flight ranging extraction:

$$R = \frac{c\Delta t}{2n_g}$$

Step 3
Weighted residual calculation for hypothesis testing:

$$J(\boldsymbol{m}) = (\boldsymbol{y} - \hat{\boldsymbol{y}}_m)^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{y} - \hat{\boldsymbol{y}}_m)$$

- The engine tests $H = \{H_0, H_{metal}, H_{void}, H_{soil}\}$ against calibration standards.
- The candidate with the lowest residual $J(m)$ is selected, explicitly capturing correlated measurement errors via the covariance matrix Σ.

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Resolving multisensor data via an evidence dependency graph

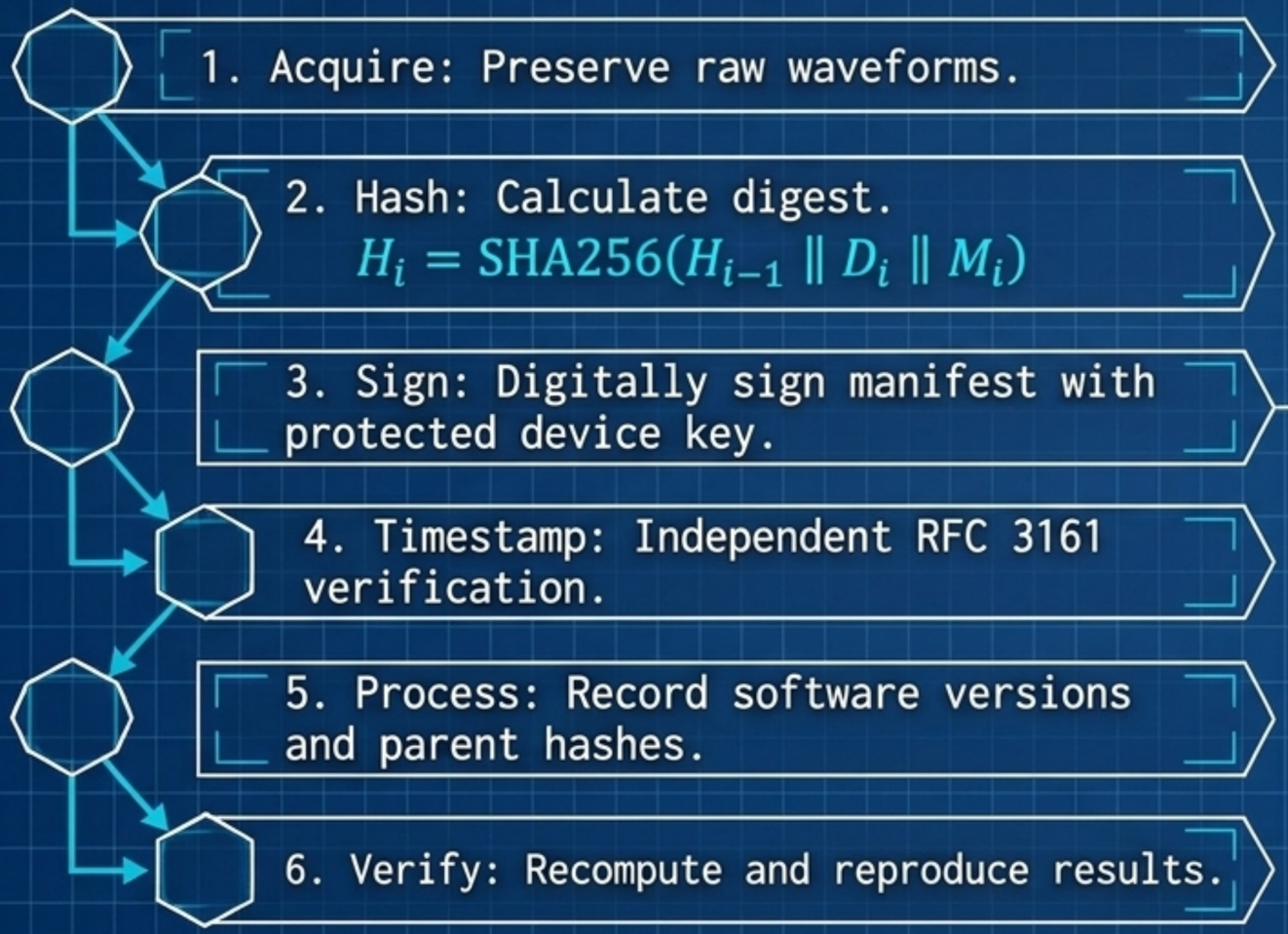
$$P(H_i \mid D) = \frac{P(D \mid H_i)P(H_i)}{\sum_j P(D \mid H_j)P(H_j)]}$$



- If three algorithms detect the same feature in one historical GPR recording, the graph forces them into a single dependency group.
- Prevents naive double-counting of a single physical anomaly.

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The Cryptographic Empirical Physics Ledger

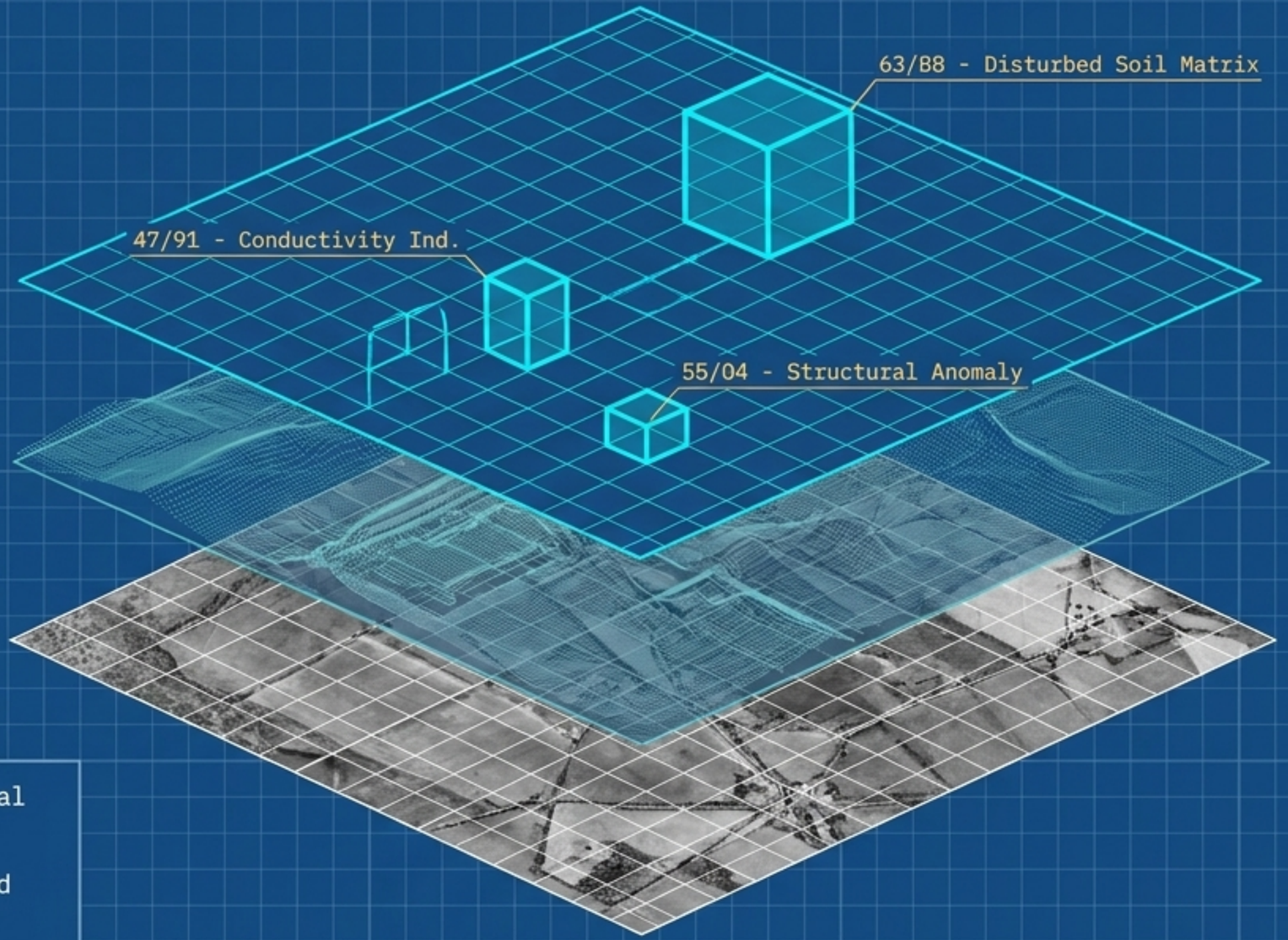


1. Acquire: Preserve raw waveforms.
2. Hash: Calculate digest.
   $H_i = \mathrm{SHA256}(H_{i-1} \parallel D_i \parallel M_i)$
3. Sign: Digitally sign manifest with protected device key.
4. Timestamp: Independent RFC 3161 verification.
5. Process: Record software versions and parent hashes.
6. Verify: Recompute and reproduce results.

- Implements a hash-linked, digitally signed, append-only evidence log.
- Guarantees that a particular signed physical record has not been altered without detection post-acquisition.

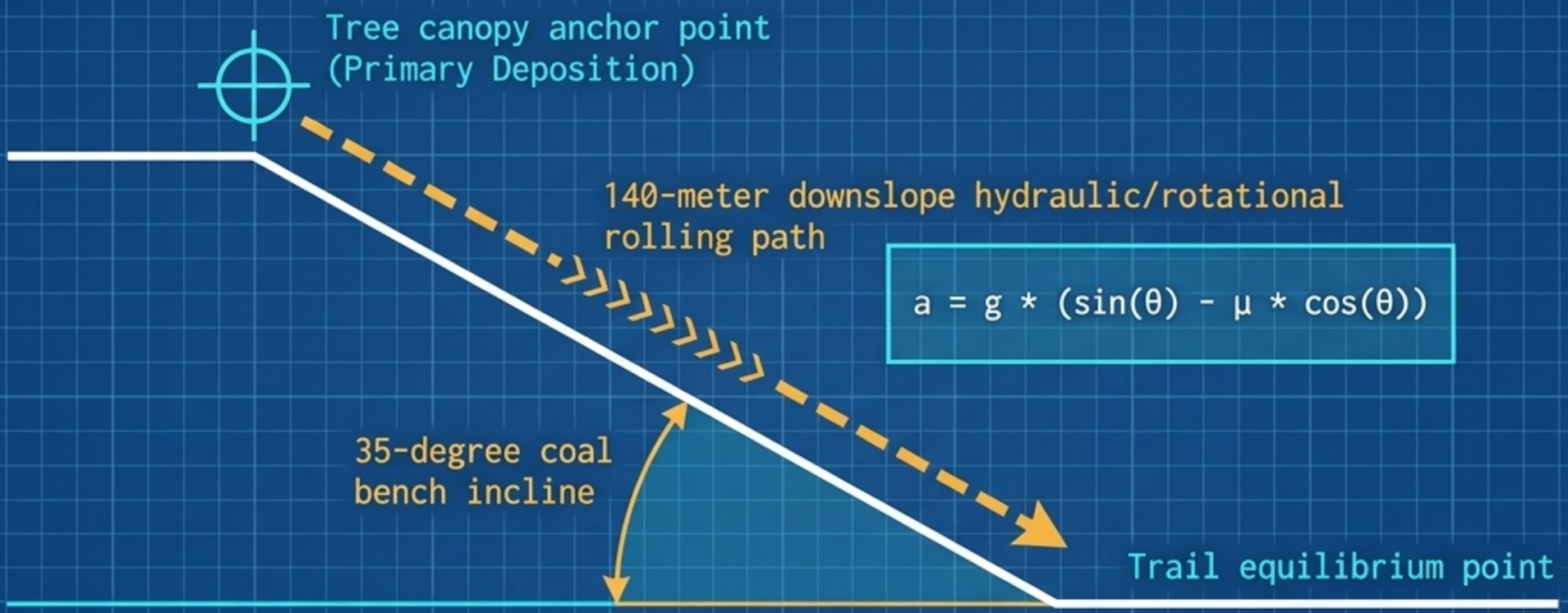
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Historical Forensic Material Reanalysis (HFMR)



- Application of the engine to historical SAR, GPR, and LiDAR.
- Extracts structural anomalies, conductivity indicators, and disturbed terrain from archival public records prior to physical field deployment.

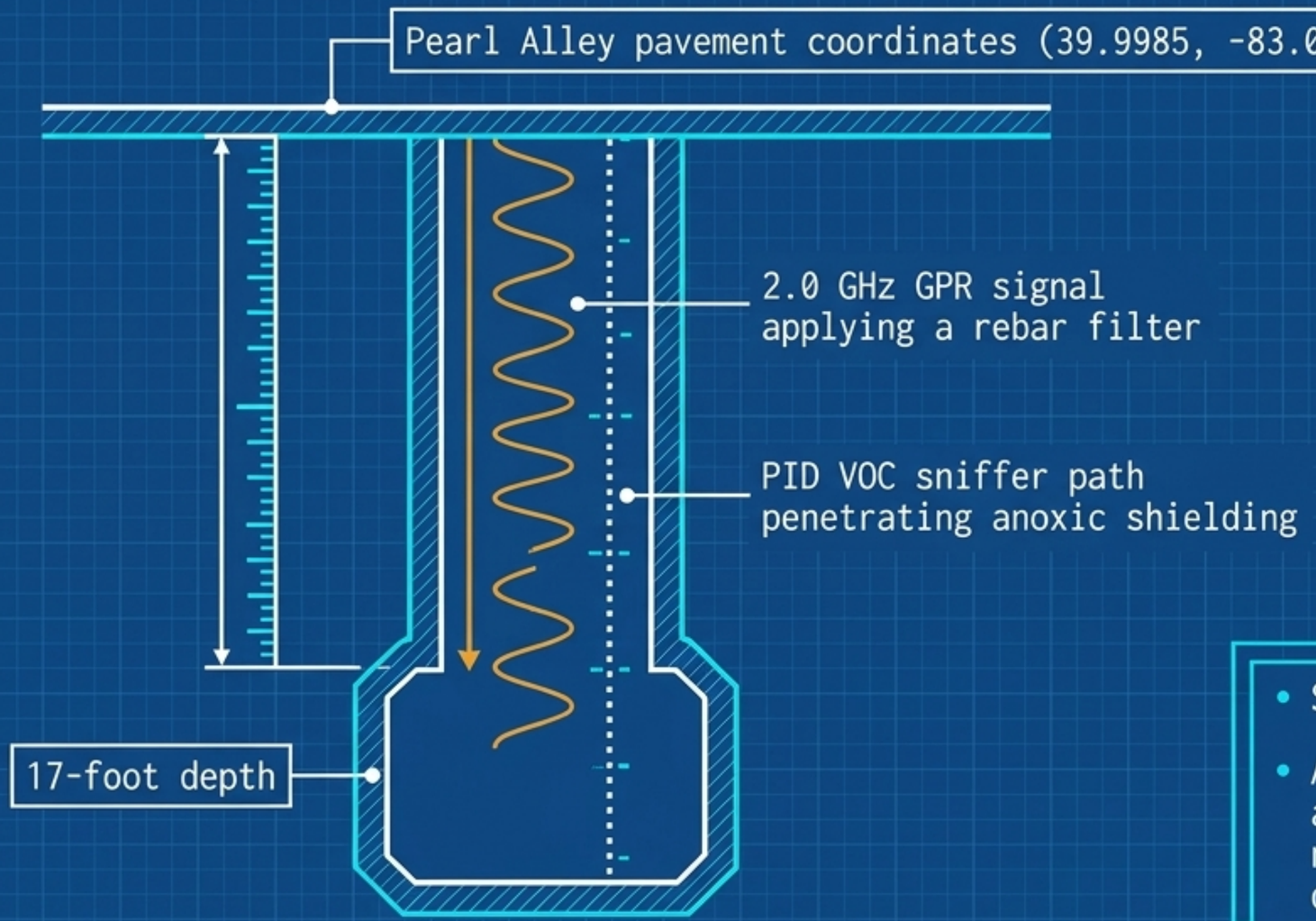
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- Deterministic geomorphology module calculates the final velocity and travel time of an object over loose matrix (friction coefficient 0.45).
- Proves natural gravitational displacement over an 11-year timeline, generating a highly precise taphonomic catchment profile.

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Subsurface Void & Attenuation Targeting (SHAFFER-2006-OH)



- Sub-pavement void targeting matrix.
- Applies deterministic dielectric attenuation formulas to calculate exact nano-second two-way travel time through dense urban shielding matrices.

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PROVISIONAL APP. / DOCKET SEC. 13: PROVISIONAL CLAIMS (SYSTEM)

Drafted Intellectual Property System Claims

1. A system for secure empirical physics metrology comprising:

(a) a data ingestion module configured to receive physical waveform data;

(b) a deterministic physics inversion engine configured to apply Maxwell-based forward models to test a plurality of material hypotheses;

(c) a cryptographic ledger configured to anchor the weighted residual output to an immutable hash chain.

Claim 1 Strength:

Asserts ownership over the data transformation and verification pipeline independent of the sensor hardware or flight apparatus.

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Drafted Intellectual Property Method Claims

10. A method for cryptographically securing forensic material analysis, the method comprising:



ingesting unclassified spatial and electromagnetic measurements;

calculating a primary SHA-256 digest of the raw acquisition matrix;

generating simulated expected sensor responses based on calibrated material standards;

calculating a multivariate covariance matrix to reject dependent observations;

writing the resulting lowest-residual material identification to an append-only directed acyclic graph.

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PROVISIONAL APP. / DOCKET SEC.15: EXECUTION & FILING

Immediate Next Steps for Micro-Entity Patent Registration

- [X] Step 1: Document Finalization. This presentation serves as the complete technical specification, background, and drawings.
- [ ] Step 2: PDF Export. Export this deck strictly as a flattened PDF to preserve monospaced formulas and structural formatting.
- [ ] Step 3: Cover Sheet Generation. Append standard USPTO Form PTO/SB/16 (Provisional Cover Sheet).
- [ ] Step 4: EFS-Web Upload. Submit as a Micro-Entity via the USPTO portal ($75 fee). Retain the resulting application number for the cryptographic ledger genesis block.

PHYS-MET

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